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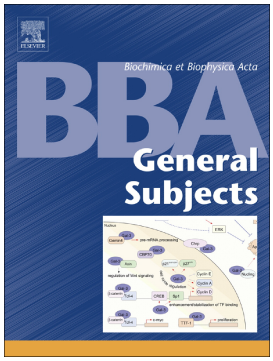

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Coating of magnetic nanoparticles affects their interactions with model cell membranes

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ABSTRACT

Background. The use of functionalized iron oxide nanoparticles of various chemical properties and architectures offers a new promising direction in gene delivery and silencing, cellular imaging and cancer imaging and treatment. The increasing applications of nanoparticles in medicine require that these engineered nanomaterials will come in contact with human cells without damaging essential tissues. Thus, efficient delivery must be achieved while minimizing cytotoxicity during passage through cell membranes to reach intracellular target compartments. **Methods.** Differential Scanning Calorimetry (DSC) and atomistic Molecular Dynamics (MD) simulations were performed for two magnetite nanoparticles with different coatings, in order to assess their different interactions with model dipalmitoylphosphatidylcholine (DPPC) membranes. The magnetite core was functionalized with polyvinyl alcohol (PVA) and polyarabic acid (ARA). **Results.** DSC experiments showed that both nanoparticles interact with the DPPC lipid bilayer, albeit to a different degree, which was further verified and quantified by MD simulations. The two model systems were simulated, and dynamical and structural properties were monitored. **Conclusions.** Our findings demonstrate that functionalized magnetite nanoparticles interact strongly with the DPPC lipid bilayers and that the interactions of the nanoparticles are strongly related to their surface coatings. DSC in combination with MD simulations is a promising strategy to assess the physicochemical interactions of coated nanoparticles with lipid bilayers. **General significance.** Understanding the physico-chemical interactions of different nanoparticle coatings in contact with model cell membranes is the first step for assessing toxic response and could lead to predictive models for estimating toxicity. Our findings demonstrate that the external coating is a key factor for identifying promising nanoparticle platforms for theranostic applications.

[1] Equal Author Contribution

Keywords: Cancer; Nanoparticles; Drug delivery; Magnetite; Molecular Dynamics simulations; Differential Scanning Calorimetry

Abbreviations

ARA polyarabic acid
DPPC dipalmitoylphosphatidylcholine
DSC Differential scanning calorimetry
MAG-ARA polyarabic acid-coated magnetite nanoparticle
MAG-PVA polyvinyl alcohol-coated magnetite nanoparticle
MD Molecular dynamics
CG Coarse-Grained
PVA polyvinyl alcohol
T_m main transition temperature
T_s pre-transition temperature
DPPC 1,2-dipalmitoyl-*sn*-glycero-3-phosphocholine

1. Introduction

In medical applications, engineered nanostructures are required to contact target cells without damaging essential tissues. The capacity of nanoparticles to reach intracellular compartments without side effects depends on their morphology and surface chemistry as well as on environmental factors such as cell type, pH, etc. Therefore, elucidating the interaction mechanisms between nanoparticles and cell membranes is of critical importance for their use in medical applications [1-5]. A number of experimental [6-11] and simulation studies [12-19] have recently focused on the effect of nanoparticle physicochemical properties on their interaction with membranes and other liquid/liquid interfaces. For example, experimental investigations of magnetite nanoparticles coated with polyarabic acid (MAG-ARA) and functionalized with doxorubicin showed outstanding membrane penetration properties, excellent drug loading and release behavior, minimal *in vivo* toxicity and promising therapeutic potential while exhibiting comparable contrasting properties with commercial agents in a clinical MRI system [20]. Surface coatings of nanoparticles also affect interfacial nanoparticle aggregation [21-23]. However, the mechanism of how the surface coating of magnetite nanoparticles can affect their cell uptake is not yet understood. In order to elucidate the different physicochemical interactions that govern the nano-bio interface, a synergy between simulation and experiments is needed.

Differential Scanning Calorimetry (DSC) can be used for monitoring the self-assembly of nanostructured materials and also for studying the interactions of nanoparticles with liposomes and lipid bilayers [24, 25]. For example, changes in the pre-transition temperature (T_s) associated with gel to rippled-gel phase and the main transition temperature (T_m) associated with rippled-gel to liquid crystalline phase for phospholipids, components of liposomes, alterations of lipids physical properties in

the presence of nanoparticles that can affect the stability of the liposome-nanoparticle assemblies may be studied with DSC [21, 23, 26-28]. From the physicochemical and biophysical point of view, DSC can easily quantify the interactions of "soft matter" (i.e., liposomes and lipids) with "hard matter" (i.e., silver and gold nanoparticles) [21, 23, 26-28]. These interactions are an indication of the potential nanotoxicity of nanoparticles as well as the loading properties of lipid carriers. Namely, changes caused by the presence of nanoparticles into liposomal bilayers may increase the fluidity of alkyl chains of lipids [23] or modify other structural and biophysical properties of lipid bilayers [21]. For example, in the case of hydrophobic nanoparticles, nanoparticles can modify the permeability of lipid bilayers, which is very often exploited for the controlled drug release properties of liposomal vectors [21-23].

Recent advancements in computer science and computational modeling have enabled the realistic simulation of key steps in the nanomedicine pipeline such as drug encapsulation and release [29, 30], nanoparticle targeting, delivery and uptake [31-36], and nanoparticle effects on cells and tissues [37-45]. Moreover, computational models of hyperthermia have been widely used to study magnetite (Fe_3O_4), maghemite (g-Fe_2O_3), and gold nanomaterials of various shapes [46-50]. Computational methodologies for modeling nanoparticles can be classified depending on the time- and length-scale they apply. These methods vary from quantum mechanical calculations to classical microscopic (atomistic) simulations and mesoscopic Coarse-Grained (CG) techniques, up to macroscopic approaches, where systems are modeled in the continuum description. Due to the size of nanoparticle-membrane systems, simulation studies of nanoparticle-membrane interactions are commonly performed with CG models [51, 52], where a group of atoms is represented by one effective interaction bead. In study cases where the atomic level of detail is needed compared to a more coarse description, atomistic molecular modeling and simulations such as Monte Carlo [53-55] and Molecular Dynamics (MD) [56-58] can provide a representation of the system macroscopic properties and direct comparison with experimental data.

Previous nanoparticle-membrane simulation interaction studies have focused mainly on gold nanoparticles [51, 59-62], carbon nanoparticles [63-66], and polystyrene nanoparticles [67-69]. These studies investigated the thermodynamics of membrane permeation of nanoparticles as well as the structural and dynamical alterations of the lipid bilayer along the process [70-72]. One of the hurdles in performing nanoparticle simulations in atomic-level detail is the actual creation of the nanoparticle crystal habit, which is required for accurate multi-scale modeling of crystal nanoparticles [73-75]. The Wulff construction can be used in order to predict the nanoparticle equilibrium shape [76], however, crystallographic files only contain the crystallographic unit cell and not its crystal habit. Therefore, creating the initial starting conformation of a nanocrystal for modeling and simulation is tedious because there is no easy way to construct the Wulff morphology from the crystallographic unit cell.

In this study, magnetite nanoparticles based on the Wulff construction and coated with the biocompatible polymers polyvinyl alcohol (PVA) and polyarabic acid (ARA) were modeled in order to investigate the cell membrane-nanoparticle interactions

and the effect of the surface coating. Because of the intricate nature of the interaction mechanism between the nanoparticle coating (ARA and PVA) and the cell membrane, atomistic (and not CG) MD simulations were employed in the presence of a DPPC model membrane. Differential interactions of the nanoparticles with the model cell membrane were monitored and are discussed below. Furthermore, differential scanning calorimetry (DSC) was also used in order to quantify the interactions of these nanoparticles with DPPC lipid bilayers. Several weight ratios of each nanoparticle were used in order to examine in-depth the effect of the coating (i.e., ARA or PVA) and the core on thermotropic characteristics (T_m and T_s) of DPPC model lipid bilayers.

2. Materials and methods

For modeling and constructing the functionalized magnetite (Fe_3O_4) nanoparticles conjugated with two types of ligands, the following procedure was used.

2.1 Construction of the magnetite core

Initially, the unit cell of the magnetite (Fe_3O_4) crystal was downloaded from the open crystallography database (http://www.crystallography.net/) with CIF number 1011032 [77]. To model the magnetite core, the Wulff crystal morphology [76] (crystal habit) of magnetite was constructed using the web-based crystallographic tool NanoCrystal, http://nanocrystal.vi-seem.eu/ [78] and the lowest surface energies for each Miller index reported in Ref.[79].

2.2 Functionalization of the magnetite core with polyvinyl alcohol (PVA) and polyarabic acid (ARA)

Subsequently, the magnetite core was coated with either polyarabic acid (MAG-ARA) or polyvinyl alcohol (MAG-PVA) using an in-house code, and the topology and parameter files for the system were constructed using the available literature. For functionalizing the magnetite core, two types of ligands are constructed, PVA and ARA. Initially, the vinyl alcohol monomer was constructed and replicated 28 times resulting in a single PVA chain of 7.5 nm length. The PVA-chain was replicated 64 times on the surface of the nanoparticle, leading to a grafting density of 3 chains/nm^2. The oxygen of the first monomer of every chain was covalently bonded to the surface of the magnetite core (MAG-PVA) to ensure a stable nanoparticle construction. The next step was the construction of the arabic acid monomer. ARA is a polysaccharide that consists of L-arabinose, D-galactose, L-rhamnose, and D-glucuronic acid in ratio 3:3:1:1. This sequence of sugars was constructed and replicated once, resulting in a double ARA chain of 6.5 nm length. The ARA-chain was considered to be linear and was replicated 64 times on the surface of the nanoparticle resulting in a grafting density of 3 chains/nm^2. The oxygen of the carboxyl group of glucuronic acid was covalently bonded to the surface of magnetite (MAG-ARA). The value of pKa of glucuronic acid is 2.83 [80]; thus, the ratio [$C_6H_9O_7^-$]/[$C_6H_{10}O_7$] is ≈31.6 in pH=7.4. Glucuronic acid is not expected to be protonated by the lipid bilayer, the metallic core or the surrounding water, at least not significantly, in this pH, hence, glucuronic acid was considered to be deprotonated throughout the

simulation. Based on these parameters, the ratio of these coated nanoparticles and the lipid bilayer is ~8:1 (w/w) for DPPC:MAG-PVA and ~4:1 (w/w) for the DPPC:MAG-ARA.

For modeling we used the CHARMM36 force field [81-83]. Parameters for the oxygen atoms of the glycosidic linkages 1-1, 1-3, and 1-4 that exist in the ARA chain were taken from Ref. [84]. The functionalized nanoparticles were then minimized and equilibrated in vacuum and water (details on section 2.5).

2.3 Construction of the lipid bilayer

A homogeneous dipalmitoylphosphatidylcholine (DPPC) lipid bilayer was constructed using the CHARMM-GUI Membrane Builder platform [85]. A rectangular box type was selected in the XY plane, and 635 lipids were placed on each leaflet (1,270 DPPC lipid molecules in total). The concentration (C) of Na^+ and Cl^- was set to 0.15 M (physiological conditions). The numbers of NaCl ions (N_{Na} and N_{Cl}) are determined by the ion-accessible volume (V) and the total charge of the system (Q_{sys}), i.e., $N_{Na} = C \cdot V - Q_{sys}/2$ and $N_{Cl} = C \cdot V + Q_{sys}/2$, to neutralize the system charge. The initial configuration of ions was then determined through Monte Carlo simulations using scaled Coulombic and van der Waals interactions [86]. The total size of the system, including water molecules and ions, was 1,091,934 atoms. The DPPC lipid bilayer was then equilibrated (details on section 2.5).

2.4 Insertion of the nanoparticles in the bilayer

After the model membrane was equilibrated, the two functionalized nanoparticles were placed in the water phase of the lipid bilayer, approximately 20 Å away from membrane surface using the visual molecular dynamics (VMD) software [87] and GROMACS tools. Overlapping water molecules and ions were removed using VMD by selecting and discarding all atoms within 2 Å of the functionalized nanoparticle. The final system was neutralized with the addition of chlorine ions to counteract charges. Finally, the two systems were minimized and equilibrated and subsequently underwent a production run of 500 ns (details on section 2.5).

2.5 MD simulations protocol

MD simulations were initially performed for the isolated systems of the magnetite core alone, DPPC/water, PVA-coated magnetite nanoparticle (MAG-PVA) in vacuum and in water, polyarabic-coated magnetite nanoparticle (MAG-ARA) in vacuum and in water, in order to equilibrate these systems before assembling the final systems of MAG-PVA:DPPC and the MAG-ARA:DPPC. All simulations were performed using the GROMACS simulation package, version 5.1.4 and the all-atom additive force field CHARMM36 [88-91]. The Particle Mesh Ewald summation technique was employed to calculate electrostatic contributions [92, 93]. The non-bonded potential energy functions were cut off at 11 Å for the Coulombic interactions and cut off and

switched at 11 Å, smoothly decaying between 8 and 11 Å for van der Waals forces. For Fe^{2+}/Fe^{3+} atoms of the magnetite core, the van der Waals parameters were retrieved from Ref. [94], and the partial charges were taken from Ref. [95]. To ensure stability for the magnetite core, harmonic restraints were applied with force constants of K_b=9,000 kJ mol^{-1} nm^{-2} and K_ϑ=9,000 kJ mol^{-1} rad^{-2} for the bonds and angles, respectively. Periodic boundary conditions were applied and the equations of motion were integrated using the leap-frog algorithm with a time step of 1 fs. The energy of the magnetite core, MAG-PVA, MAG-ARA, MAG-PVA-DPPC, and MAG-ARA-DPPC systems were minimized using the Steepest Descent algorithm for 50,000 steps.

The magnetite core, MAG-PVA and MAG-ARA systems were equilibrated for 100 ps in vacuum under a constant number of particles, volume and temperature (NVT ensemble) for 100 ps using the Velocity rescale thermostat [96], with a reference temperature of 310 K and time constant of 0.1 ps. Then, the MAG-PVA and MAG-ARA systems were simulated for another 10 ns, until sufficient folding of polymer chains around the magnetite core was achieved. Afterwards, the systems were solvated using the TIP3P water model [97] and the magnetite core was minimized again. Equilibration followed under a constant number of particles, pressure, and temperature (NPT ensemble) for 100 ps for the MAG-PVA and MAG-ARA systems and for 500 ps for the magnetite core. The temperature was kept constant at 310 K using the Nose-Hoover thermostat with a relaxation time of 1 ps [98]. The pressure of the system was isotropically coupled and maintained at 1.01325 bar using the Parrinello-Rahman algorithm with a time constant of 5 ps and compressibility of 4.5e-5 bar^{-1} [99]. A production run of 10 ns followed for the MAG-PVA and MAG-ARA systems in water.

The DPPC lipid bilayer was simulated for 225 ps gradually equilibrating temperature, pressure, and membrane area. The temperature was kept constant at 323.15 K using the Berendsen thermostat with a relaxation time of 1 ps [100]. The pressure of the system was semi-isotropically coupled and maintained at 1.01325 bar using the Berendsen algorithm with a time constant of 5 ps and compressibility of 4.5e-5 bar^{-1}. The next step was to extend the simulation for 100 ns, ensuring that the model membrane is stable using the Nose-Hoover thermostat and the Parrinello-Rahman barostat with the same parameters as previously mentioned.

For the MAG-PVA:DPPC and MAG-ARA:DPPC systems, the temperature was kept constant at 323.15 K using the Velocity-rescale thermostat with a relaxation time of 0.1 ps. Initially, the Parrinello Rahman barostat was employed in order to stabilize the pressure of the system. This procedure did not work since the volume of the simulation box started to oscillate rapidly, and the system disassembled. Thus the pressure of the system was relaxed in two discrete steps. Firstly, the Berendsen algorithm was employed (300 ps) with a time constant of 5 ps and compressibility of 4.5e-5 bar^{-1} being stable, but not reproducing the proper ensemble (NPT). Therefore, in the second step, the Parrinello Rahman barostat was used (300 ps) with a time constant of 5 ps and compressibility of 4.5e-5 bar^{-1}. Pressure and temperature converged to their desired values of 1.01325 bar, and 323.15 K respectively. Finally, the two fully equilibrated systems underwent a production run of 500 ns, and the results were analyzed to study the positioning of the nanoparticle in the bilayer, and the interactions of ARA and PVA in atomic-level detail.

2.6 Differential scanning calorimetry

We used DPPC phospholipids to prepare model lipid bilayers. Lipids were purchased from Avanti Polar Lipids Inc., (Albaster, AL, USA) and used without further purification. MAG-ARA and MAG-PVA nanoparticles were bought from Chemicell in aqueous dispersion. The protocols used for the thermodynamic characterization of DPPC bilayers with nanoparticles are described in a previous report [20] and in detail in the Supporting Information. As the simulations with membranes were performed at 323.15K, which is above the gel-to-liquid transition temperature for DPPC membranes (Tm = 314.15K). For this reason, while the DSC experiments were carried out at temperatures lower and higher than T_m. Namely, three heating-cooling cycles were performed: 10° to 60°C, at 2°C/min or 20°C/min scanning rate, respectively. The thermodynamic evaluation was performed during the second cooling and heating cycle.

3. Results

3.1 Construction and equilibration of functionalized magnetite nanoparticles and and model membrane

First, the Fe_3O_4 nanoparticle was obtained using a radius of 1.5nm resulting to 1.192 atoms (Fig. S1).

The magnetite core was attached to either PVA or ARA (Fig. S2) using 28 repeat units for PVA and 2 repeat units for ARA leading to a chain length of 7.5 nm for PVA and 6.5 nm for ARA and a grafting density of 3 chains/nm^2 in each system. The total number of chains on each functionalized nanoparticle is 64.

The folding of the polymer chains in water is computationally very demanding. For this reason, the functionalized nanoparticles were first simulated in vacuum for 10 ns (Fig. S3). The resulting structures were then placed in water and simulated for 10 ns (Fig. S4). Polymer chains partially unfold allowing water molecules to penetrate between them due to the hydrophilic character of the chains.

Furthermore, the stability of the model membrane was examined by means of area per lipid, which was found to be 62.46 ± 0.50 Å^2 (see Fig. S5), very close to the experimental-reported value for liquid DPPC bilayers.

3.2 MD simulations analysis of functionalized nanoparticles

Quantification of the MD trajectory, employing mean square displacement (MSD), radial distribution functions, hydrogen bond analyses and membrane dynamics showed that both nanoparticles interact with the DPPC lipid bilayer albeit to a different degree.

To assess the interactions between the functionalized nanoparticles and the model membrane, we calculated the diffusion coefficient by means of MSD. The diffusion of both systems appears to be bimodal (Figure S6). We used bootstrapping to sample over different trajectory lengths, hence, the slope of the mean square displacement was measured only up to 200ns as beyond this time scale we have poor statistics. Knowing the structure of both systems, it can be concluded that the model membrane represents a physical barrier that slows down both nanoparticles. The first mode represents the diffusion of both nanoparticles in the water phase, while the second mode emerges when both nanoparticles interact strongly with the model membrane. Linear fitting in the areas of 10-85 ns and 85-200 ns in the first case (MAG-PVA-DPPC) results in the diffusion coefficients D_{10-85} = 0.0513 ± 0.0113 10^{-5} cm^2/s & D_{85-200} = 0.0147 ± 0.0028 10^{-5} cm^2/s, respectively. In the case of MAG-ARA-DPPC, linear fitting took place in two separate regions: 10-90 ns and 90-200 ns. The two diffusion coefficients are D_{10-90} = 0.0472 ± 0.0051 10^{-5} cm^2/s and D_{90-200} = 0.0167 ± 0.0060 10^{-5} cm^2/s, respectively. Visualization of the MD trajectory for the MAG-ARA:DPPC system shows that the arabinose groups of the ARA interact with the choline and phosphate groups of the DPPC bilayer. In the case of MAG-PVA:DPPC system, the MD trajectory clearly shows distinctly that the vinyl alcohol groups interact strongly with the lipid bilayer and such interactions can be characterized equivalent to those of MAG-ARA/DPPC due to similar MSD values of the MAG-ARA system shown in Fig. 1 and Fig. S6. Subsequently, the solvation of the two functionalized nanoparticles was assessed through radial distribution functions. In Fig. S7 it can be seen that both systems show the same solvation behavior.

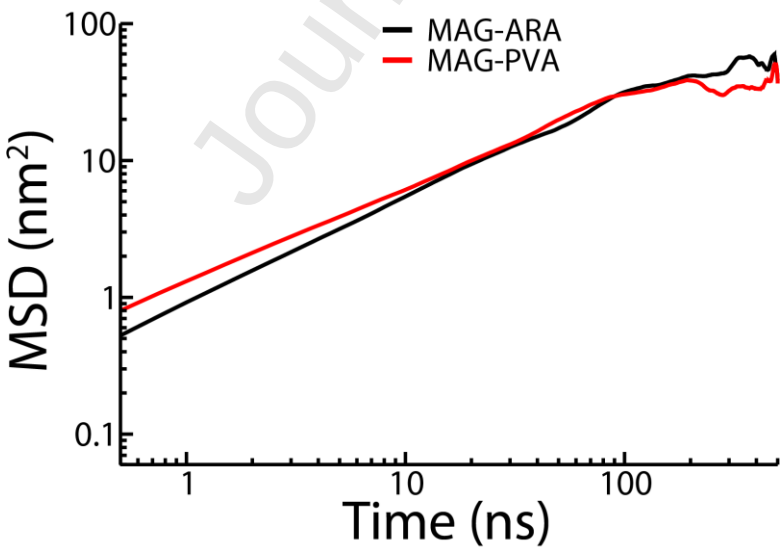

Figure 1. Diffusion of MAG-PVA and MAG-ARA.

The first coordination number, which is derived from the area under the first peak of *g(r)*, was assessed through integration at the first minimum of each curve. For the

pairs MAG-ARA/water and MAG-PVA/water at x_{min}=2.04 Å, the first coordination number is $n_{1,\ MAG\text{-}ARA/Water}$=0.05088 & $n_{1,\ MAG\text{-}PVA/Water}$=0.05130 respectively. The same procedure was followed for the magnetite oxygen atoms/ARA oxygen atoms and magnetite oxygen atoms/PVA oxygen atoms, as depicted in Fig. S8. For the pairs MAG oxygens/ARA oxygens and MAG oxygens /PVA oxygens at x_{min}=3.42 Å & x_{min}=3.44 Å, respectively, the first coordination number is $n_{1,MAG\ oxygen/ARA\ oxygen}$=36.783 & $n_{1,MAG\ oxygen/PVA\ oxygen}$ =14.406, respectively. Thus, it can be deduced that the ARA coating has a stronger interaction with the magnetite metallic core.

The interactions between the functionalized nanoparticles and the model membrane were also analyzed utilizing the radial distribution function (Fig. 2). It is evident that the coordination number of the PVA oxygen atoms from the DPPC nitrogen or phosphorus atoms is higher than that of ARA oxygen atoms from DPPC nitrogen or phosphorus atoms, for every distance in the calculated distance range. Thus, it can be deduced that the MAG-PVA system interacts more favorably with the model membrane than the MAG-ARA system.

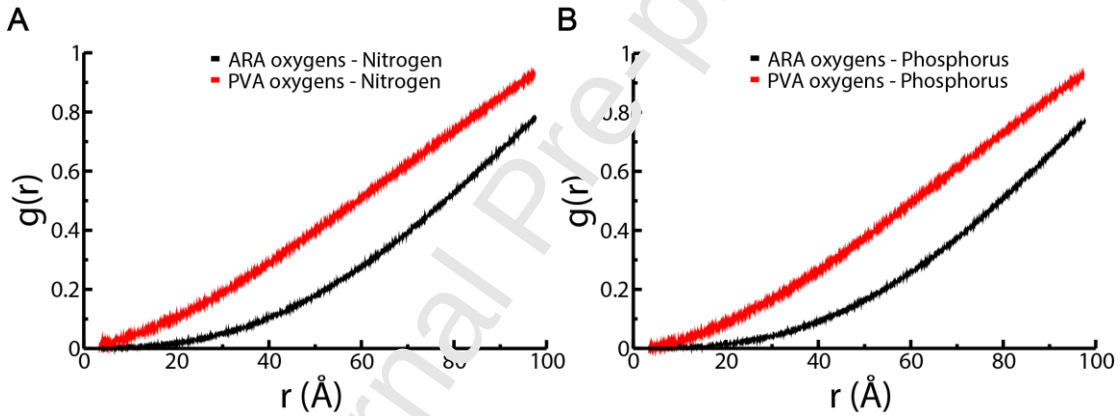

Figure 2. a) Radial distribution function between ARA oxygen atoms – DPPC nitrogen atoms and PVA oxygen atoms - DPPC nitrogen atoms b) ARA oxygen atoms – DPPC phosphorus atoms and PVA oxygen atoms - DPPC phosphorus atoms.

Interactions between the nanoparticles and the lipid bilayer were also evaluated via hydrogen bond analysis. Using the built-in module of Gromacs the total number of hydrogen bonds between the nanoparticles and model membrane were quantified as a function of time, as well the number of hydrogen bonds as a function of the donor-acceptor distance (Fig. 3).

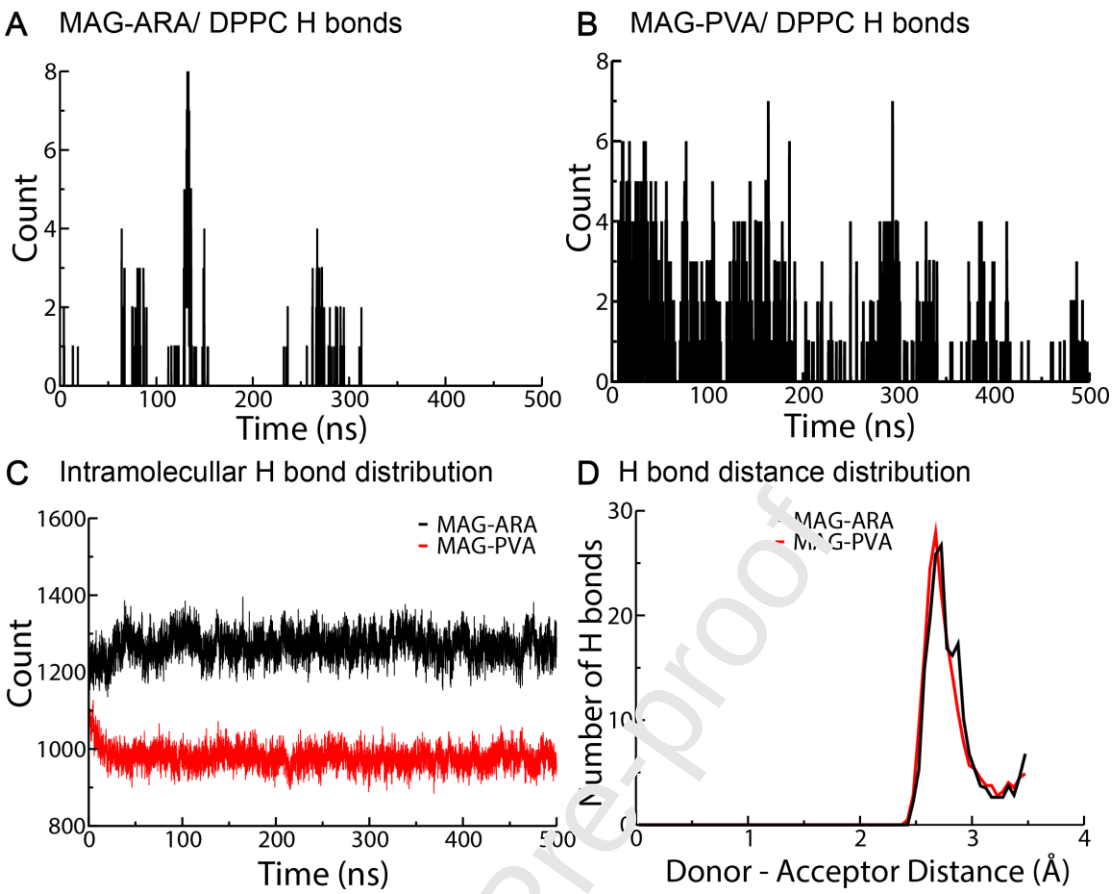

Figure 3. Hydrogen bonds to quantify the interactions between nanoparticles and model membrane. (A) Number of hydrogen bonds as a function of time for the MAG-ARA / bilayer system. (B) Number of hydrogen bonds as a function of time for the MAG-PVA / bilayer system. (C) Number of intramolecular hydrogen bonds for the MAG-ARA and MAG-PVA (red). (D) Distance distribution of the hydrogen bonds between MAG-ARA/bilayer and MAG-PVA/bilayer.

The analysis reveals that the MAG-PVA nanoparticle interacts with the lipid bilayer much more frequently than the MAG-ARA one (Fig. 3A, B). In the case of the MAG-PVA the maximum number of simultaneous hydrogen bond interactions are 7 and the time series shows the interactions are much more frequent. In the case of MAG-ARA the interactions with the bilayer are present, but appear sporadically through the trajectory. Nevertheless, the highest number of simultaneous interactions are found to be 8, compared to 7 found in MAG-PVA. The trajectory shows that in both cases the nanoparticles anchor into the lipid bilayer when interacting (Fig. 4). A reason for the above could be due to the intramolecular interactions in the form of hydrogen bonds for the two systems discussed. We computed the total intramolecular hydrogen bonds for the MAG-PVA nanoparticle to be 979 ± 27 and for the MAG-ARA, 1268 ± 35 (Figure 4C). Consequently, the MAG-ARA intramolecular interactions with the magnetite metallic core could be a reason why it shows less contacts with the bilayer as compared to MAG-PVA, since they both show an almost identical profile for the donor acceptor distance (Figure 4D).

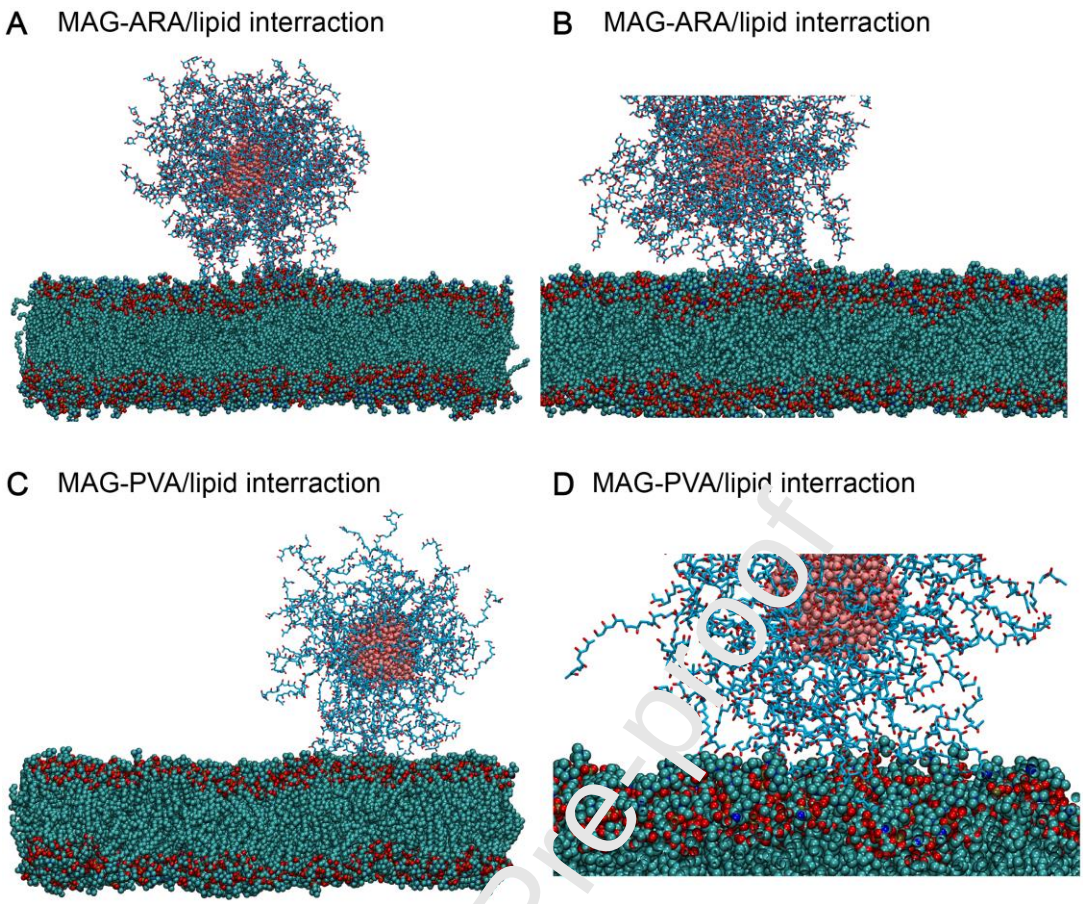

Figure 4. Nanoparticles interacting with the lipid bilayer. (A, B) MAG-ARA nanoparticle interacting with the lipid bilayer. (C, D) MAG-PVA nanoparticle interacting with the lipid bilayer.

From the quantification of nanoparticle/bilayer interactions via hydrogen bonds (Fig. 3) and visual inspection (Fig. 4), the question was raised, whether the dynamics of the bilayer are affected. Indeed, the bilayer responds to the presence of the nanoparticles. We computed the area per lipid from a well-equilibrated part of the trajectories (last 50ns) (Fig. 5). Whilst both time series of the area per lipid are fairly similar in value, we find that the overall values are reduced compared to the native bilayer (Fig. S5) meaning that both nanoparticles induce a rigidification of the bilayer with MAG-PVA exhibiting a stronger effect. In addition, the distribution of the area per lipid values in the case of MAG-PVA (Fig. 5D), are more centered around the central value as compared to the MAG-ARA system (Fig 5C). Examination of the time series of the bilayer thickness also shows that the bilayer thickness is also slightly increased for the MAG-PVA system consistent with a more rigid bilayer core (Fig. S9). The hydrodynamic radii of the nanoparticles were not affected by the interaction with the bilayer and both were stable at 120 Å (Fig. S10).

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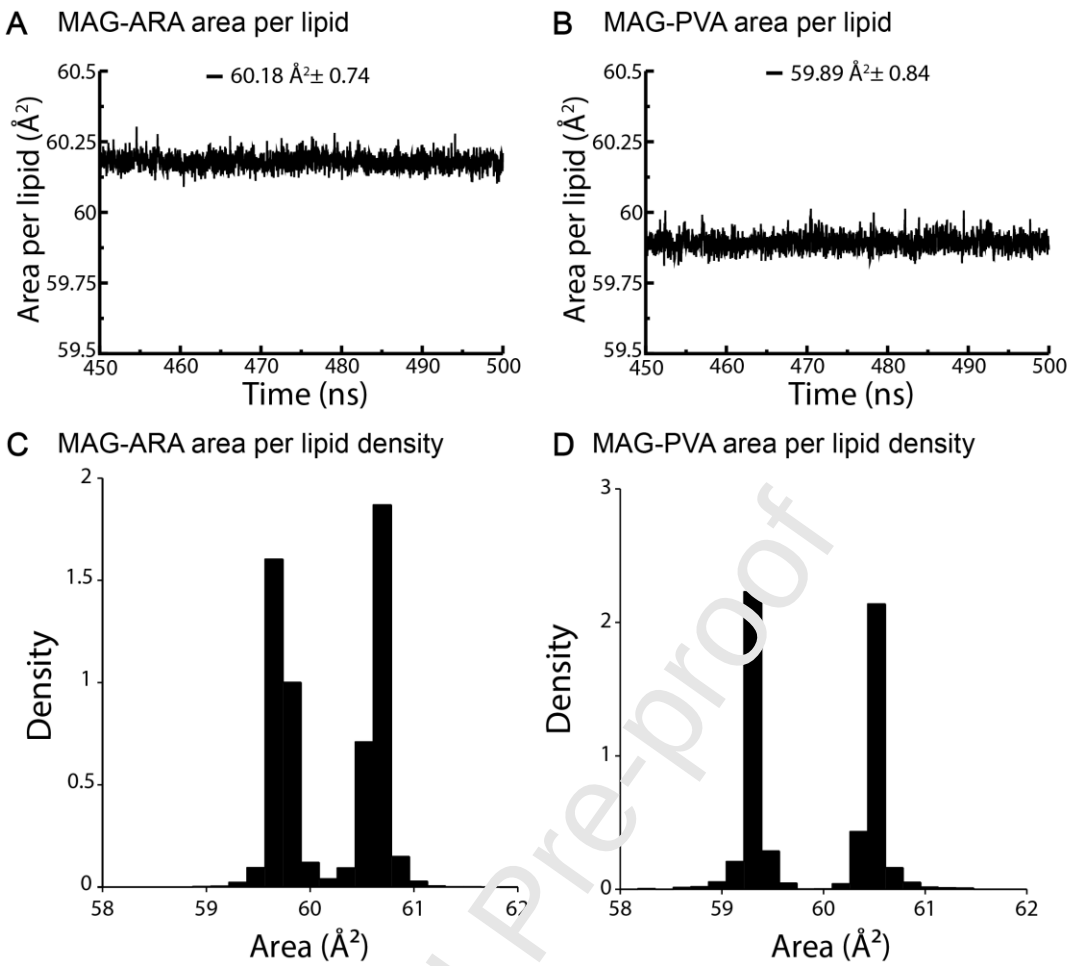

Figure 5. Nanoparticles affect bilayer dynamics. (A) Area per lipid for the DPPC bilayer in the MAG-ARA system. (B) Area per lipid for the DPPC bilayer in the MAG-PVA system. (C) Area per lipid density for the DPPC bilayer in the MAG-ARA system. (D) Area per lipid density for the DPPC bilayer in the MAG-PVA system.

To monitor the relative movements of the nanoparticles to the bilayer, we plotted the distance between magnetite core center of mass and the bilayer center of mass versus time. Figure 6 shows that MAG-ARA fluctuates on average further away from the bilayer core with an average distance of 10.6 ± 1.3 nm , while MAG-PVA is at 8.6 ± 1.2 nm. The fact that the hydrodynamic diameter of the two nanoparticles remains the same at 12 nm means that the MAG-PVA is indeed interacting more closely with the DPPC bilayer.

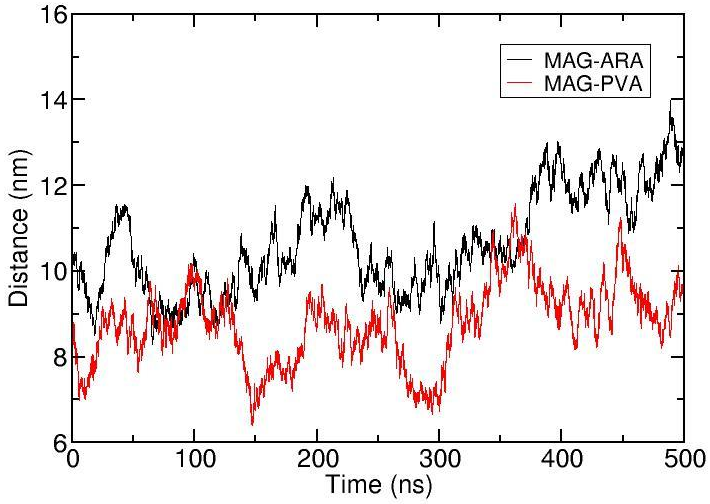

Figure 6. Distance between magnetite core center of mass and the bilayer center of mass versus time. In black, MAG-ARA is shown and in red MAG-PVA.

3.5 Interactions of MAG-PVA and MAG-ARA nanoparticles with DPPC membrane models using DSC

Model lipid bilayers are usually used to investigate the interactions of active substances, biomacromolecules (i.e., polymers, dendrimers, etc.) and nanoparticles with biological membranes [101-103]. In this study, we used DPPC hydrated lipid membranes for simulating the biological membrane. For DPPC, the main transition temperature (T_m) is 41.5°C and the pre-transition temperature (T_s) is 36.5°C [101-103]. These two thermal phenomena correspond to the melting of the hydrocarbon chains leading to the liquid crystalline phase (Lc state) and the organization of rippled phase (Pβ state), respectively. Fig. 7 presents the heating scans of DPPC:MAG-PVA (Fig. 7A) and DPPC:MAG-ARA (Fig. 7B) at different weight ratios of lipid/nanoparticles. The values of all the thermotropic parameters and the cooling scans are presented in Tables S1 and S2 and Fig. S11, respectively. For the MAG-PVA nanoparticle, we observe that the T_m remains unaffected as the weight ratio of the nanoparticle is increased (Fig. 7A and Table S1). The T_m remains at ≈ 41°C for the DPPC:MAG-PVA systems, similar to the T_m of the pure lipid, and this value is in accordance with the values observed in the literature. This result means that there are no strong interactions between MAG-PVA and the alkyl chains of the phospholipids. Nevertheless, there are interactions between the PVA (surface coating of the nanoparticle) and the polar head groups of DPPC at 100:1, 40:1 and 2:1 w/w ratios, probably due to the different conformation of the polymeric chains. The values of the ΔH_m, which correspond to the main transition and ΔH_s, which correspond to the secondary/pretransition, are increased significantly due to the above interactions. In our opinion, the conformation of the PVA plays a key role in the interactions with polar groups, and consequently, these interactions also affect the orientation of the head groups. In particular for the 40:1 concentration, our

results indicate that the penetration of the MAG-PVA nanoparticle into DPPC lipid bilayers does cause the formation of new metastable/new crystalline phases. We also observe strong interactions of MAG-ARA nanoparticles with the polar groups of DPPC phospholipids, especially at DPPC:MAG 10:1 and 2:1 w/w because the secondary transition enthalpy is decreased significantly (Table S2). The secondary transition enthalpy corresponds to the pre-transition enthalpy of the lipids and is an indication about the orientation and the interactions of polar phosphate groups with species. Namely, there is a reduction of ΔH_s as the ratio w/w increased, indicating the different hydration forces/interactions between the polysaccharide of nanoparticles and polar groups of DPPC lipid bilayer. A decrease of the T_s indicates that the interactions of nanoparticles with sugar molecules leads to a decrease in thermal energy (enthalpic effect), which is responsible for the reorientation of the lipid bilayers from rippled to solid-gel liquid crystalline phase (Fig. S11, Table S2). The cooperativity of the DPPC lipids decreased significantly with the increase of weight ratio of MAG-ARA, as indicated by the $\Delta T_{1/2}$ values (half-width at the half peak height of the transition). The thermal effect starts at a lower temperature at the highest weight ratio of the MAG-ARA, too. The same phenomena are also observed for the main transition, but these are the result of the strong interactions between the surface coating (sugar molecules) of the MAG-ARA nanoparticle with the polar group because the main transition onset and peak remain similar to those of the pure DPPC. Furthermore, the main transition enthalpy decreased due to these interactions, without any significant change of the T_m and $T_{onset,m}$, as described before. Finally, we can conclude that the interactions of the nanoparticles are strongly related to their surface coatings (i.e., PVA and ARA).

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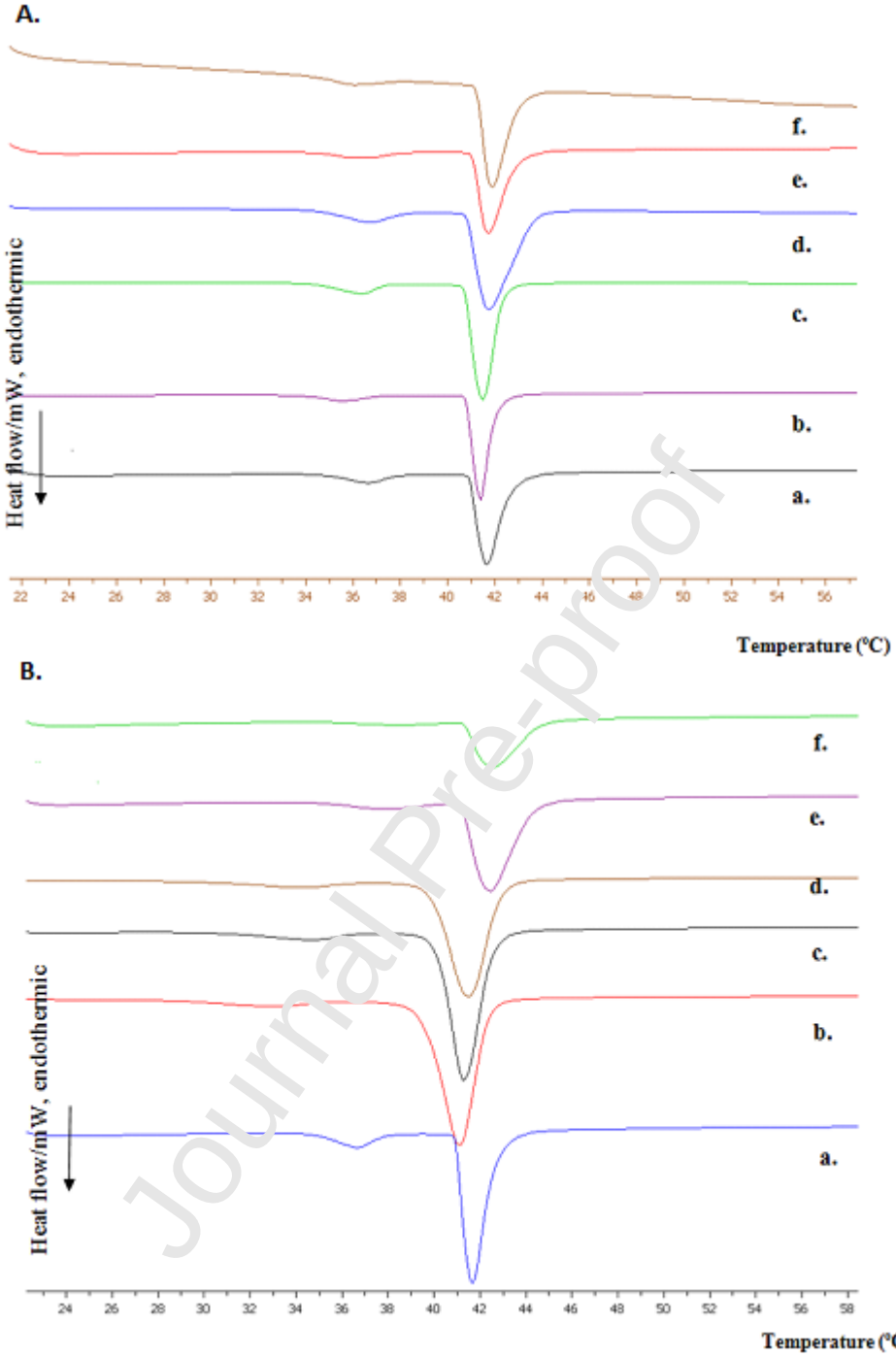

Figure 7. DSC heating scans of A. DPPC:MAG-PVA and B. DPPC:MAG-ARA at a. 1:0, b. 200:1, c. 100:1, d. 40:1, e. 10:1, and f. 2:1 weight ratios. The limits for the calculation of thermotropic parameters are from 25°C to 50°C.

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4. Discussion and Conclusions

Functionalized magnetite nanoparticles are a promising new technology that could enhance the efficiency of drug delivery carriers and diagnostics tools for novel theranostic applications. To assess the cytotoxicity of those particles, the interactions between them and the cellular membrane must be quantified. For this reason, MD simulations and DSC experiments were carried out for two magnetite nanoparticles with different coatings (PVA and ARA), and the interactions with the membrane were quantified and compared with each other.

Functionalization plays a key role in the interactions of the magnetite nanoparticles with lipid bilayers, as the obtained thermograms revealed by DSC experiments. DSC showed that there are no strong interactions between the functionalized magnetite nanoparticles and the alkyl chains of the phospholipids, as also confirmed by the simulations. For these reasons, from a thermodynamical point of view, the MAG-ARA and MAG-PVA did not cause any destabilization of the biological model membrane, which can be translated as toxicity of these nanoparticles on the biological level. MAG-PVA and MAG-ARA showed differential interactions with the DPPC polar head groups depending on their different concentrations studied. Both nanoparticles showed interactions with the DPPC head groups for the respective concentrations studied with simulations and the different observed effects can be attributed to the different coatings, which was further rationalized by MD simulations and examination of the interactions in atomic-level detail.

For the simulations, after building the topology of the magnetite core and the two coatings, suitable parameters were assigned and the cell membrane was built. Subsequently, the two systems (MAG-PVA:DPPC & MAG-ARA:DPPC) were simulated, and dynamical and structural properties were monitored. Both systems appear to have bimodal diffusion. Moreover, the overall diffusion for both systems does not differ, as shown in Fig. 1. Taking into consideration the structure of both systems, it can be deduced that the model membrane represents a physical barrier that slows down both nanoparticles and statistically significant differences cannot be extracted. Visualization of the MD trajectory shows that the nanoparticles diffuse freely in the water phase, and as they approach the lipid bilayer, they interact strongly. Moreover, by plotting RDF versus distance it was shown that MAG-PVA interacts stronger than MAG-ARA since its oxygen atoms approach closer to the polar heads of the DPPC bilayer, while MAG-ARA prefers to interact more strongly with the magnetite metallic core. Hydrogen bond analysis showed that the MAG-PVA nanoparticle interacts with the lipid bilayer much more frequently than the MAG-ARA one. Although none of the two nanoparticles enter the bilayer core within the timescale of the simulation, they both affect the total area per lipid. Overall area per DPPC values are reduced compared to the native bilayer meaning that both nanoparticles induce a rigidification of the bilayer with MAG-PVA exhibiting a stronger effect. Also, MAG-PVA is on average closer to the lipid bilayer compared to MAG-ARA.

Nanoparticles of different cores exhibit different properties and depending on the core material they also demonstrate different mechanisms of nanomaterial cytotoxicity such as oxidation via formation of reactive oxygen specials and free

radicals; damaging of cell membranes; impairment of components of the cytoskeleton; disturbing intracellular transport and cell division; damaging transcription and damage DNA; interference with essential organelles such as mitochondria and lysosomes; causing structural changes in membrane proteins and altering signal transduction.[104] For example, TiO2 NPs induce conformational changes in tubulin and inhibit its polymerization, which disturbs intracellular transport, cell division, and cell migration.[105] Here, we argue that the nanoparticle coating will alter the properties of the "naked" nanoparticle, and the functionalized coated nanoparticles will exhibit different characteristics compared to the pure nanomaterial core. Depending on the degree of coating the core will or will not interact with the lipid bilayers. With a grafting density of 3 chains/nm^2, we consider that the nanoparticle core is fully coated. As the nanoparticle does not penetrate the lipid bilayer and only interacts with water and the lipid polar head groups here the composition of the core did not affect the properties of the nanoparticle in contact with a model cell membrane.

In the present study, we performed both simulations and wet-lab experiments in order to quantify the interactions between DPPC lipid bilayers and MAG-ARA/MAG-PVA nanoparticles. By combining the two methodologies, the following outcomes could be concluded: 1. No strong interactions appeared between DPPC alkyl chains and MAG-PVA or MAG-ARA nanoparticles. The nanoparticles did not enter the lipid bilayer core in simulations, but also not in DSC experiments as evidenced by the fact that no significant change of the T_m and $T_{onset,m}$ occurred for both nanoparticles. 2. Strong interactions developed between MAG-PVA and MAG-ARA nanoparticles with polar groups of phospholipids, albeit to a different degree, depending on the surface coating. 3. No new metastable phase was created in the lipid bilayer in the presence of the MAG-ARA or MAG-PVA nanoparticles in the DSC experiments as also confirmed by the simulations. 4.The folding of the polymer chains in the water environment plays a key role on the interactions with phospholipids' polar groups. 5. Due to the fact that the MAG-ARA and MAG-PVA did not cause any destabilization of the biological model membrane it can be concluded that the aforementioned functionalized magnetite nanoparticles are promising technological platforms for theranostic purposes and biomedical applications.

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Conflict of Interest and Authorship Conformation

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

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Highlights

- Strong interactions developed between MAG-PVA and MAG-ARA nanoparticles with polar groups of phospholipids, albeit to a different degree, depending on the surface coating.
- MAG-PVA interacts more strongly with DPPC headgroups inducing rigidification of the bilayer
- MAG-ARA and MAG-PVC do not destabilize the model membrane and could be considered promising technological platforms for theranostic purposes and biomedical applications.

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